

Store Locator Software

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In
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Executive Summary of Proposal

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The purpose of the capstone project is to develop a software component that will provide an easy to use development process for Store Locator functionality that meets all the latest up to date technology in interactive maps. The goal is to minimize the development time that developers will need to spend in creating their desired functionality. In order to achieve this goal, an NPM package will be deployed that developers can download and install via NPM and use to develop in their Angular applications.

In person shopping is still a highly important avenue for businesses. This is despite the growth of online shopping and eCommerce. Businesses still need a way to get customers into stores. Customers need a way of locating their nearest stores. Many businesses are lacking in the way they provide for this need. Many do not have interactive locator maps on their sites. This project aims to solve this problem by providing a solution for developers to streamline their development of a Store Locator in their online presence.

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PART I

Background and Approach

Introduction/Background

For the proposed Capstone project, the student will be developing software to implement a store locator. The student is a Software Engineer who has worked for several years now, and has gained inspiration for this project through working with a lot of software components that have made the development process easier. The motivation for this project was to create something that was as useful for software developers during the development process.

An example of such a software component was a component for creating an HTML table that had in-built support for pagination and sorting. It even supported in place editing. These were features that were supposed to be included on the page as a software requirement, and instead of having to build them from the ground up, the student was able to leverage what was supported by the component. This saved hours, potentially days, of development time. The goal of the capstone project is to deliver something that provides the same benefit to another developer with the added bonus of being something usable by those that are less tech savvy.

According to Wandosell et al, online store locators play an important role as one of the tools available to online retailers. Many online retailers already have store locators on their websites, but not all of them are utilizing the latest in interactive map technology, and not all of them are providing all of the advanced features such as route planning that can help increase the engagement with potential customers. Based on the statistics (Wandosell et al., n.d., 8), only 40.5 percent of online retailers were utilizing the most up to date technology by providing route planning in their store locators.

This is a problem. Interactive maps and route planning are now the cutting edge of store locators. Not having these up to date features means that businesses are missing out on a lot of potential engagement with their customers. Furthermore, the number of customers who browse the web on their phone is extremely high. These users want to be able to get driving directions to stores on their phones as they are browsing. It is vitally important that businesses provide these features to their end users. This project aims to solve this problem and provide for these needs. The specific type of software solution that it will be providing is a store locator. Not only will this project be useful to developers, but it is also a project that helps solve a business need.

Project Goals and Objectives

Goal: Finalize Tech Stack to Use	Goal: Learn How to Create and Deploy NPM Packages	Goal: Complete the Final Software
Objective: Research Available Tech Stacks 1 Hour Daily	Objective: Study the NPM docs 1 hour daily until material is understood	Objective: Code the interactive Map (Google Maps API) 2-3 hours daily
Objective: Code Sample Projects Weekly --- Until Class Starts at latest	Objective: Watch tutorial videos 1 hour daily until material is understood	Objective: Code the data sources - 2-3 hours daily
	Objective: Code sample NPM packages and upload them to NPM Weekly	Objective: Code the layout of the application

Goal: Deploy final Software to NPM	Goal: Design look of the application
Objective: Test the software until satisfied	Objective: Create a mock. 4 hours.
Objective: When completed, deploy	Objective: Revise mocks as many times as needed
	Objective: Benchmark other store locator designs

Above are the goals and objectives that the student has for the project. In the short term, the student will need to identify the exact tech stack that he will be developing in. Additionally, this choice will decide which technologies that the software will support. The goal is to make the software as widely available as possible, but in the time constraints of this class the student will have to make sure to limit the scope to something that is completable in time. So the first goal that is listed is to finalize the tech stack that will be used. In order to accomplish this task, there are several daily and weekly objectives that will need attending to. The first is to research the technology, and the second is to code sample projects to verify that the skills needed to work in that technology are fully developed.

The second main goal is to learn how to create NPM packages. In order to accomplish this goal, there are three objectives to implement daily and weekly. These are to study the npm documentation for 1 hour daily until the material is understood, watch tutorial videos for an hour daily until the material is understood, and to code sample NPM packages weekly. All of these objectives are necessary for achieving the goal of learning how to create and deploy the software.

The third main goal is to design the look of the application. In order to do this, there are two objectives that have been identified: creating a mock layout, and revising the layout. It is possible that these two objectives can be broken down further. If they were to be broken down further, they would be into finding the mock software, benchmarking designs, and laying out the web page. Revising can then be broken down into the exact same sub-objectives, just repeated over and over until a final design is completed.

The fourth goal is to complete the final software. This has been broken down into three objectives: coding the layout of the application; coding the interactive map; coding data sources. Like the goal of designing the layout, this could be broken down even further. The interactive

map can be broken down into coding the markers on the map, coding the route planning, coding the geolocation, and coding the search.

The fifth and final goal is the ultimate deployment of the packages to npm. Listed are two objectives for this goal: testing and the actual deployment. This goal and its objectives are kind of blurring the line between goal and objective. The main goal of deployment is the main overarching goal of the project, the ultimate final deployment of the software. On the other hand, the actual physical action of deployment is a rather small objective, the very final act of doing the deploy to npm. In this weird way, the deployment is both a big goal and a small objective that is the very last step. It has been included in the table in this way because deployment involves testing first, and then the commands that need to be run to physically deploy the software.

Environmental Scan

Physical retail is the most used operation format around the world. (Wandosell et al., n.d., 1). Because of this fact, it remains a huge business need to get people into stores. That does not mean that online shopping is not seriously growing rapidly; it just means that a store locator is a hugely valuable resource for companies. According to Wandosell, et al, there were several main services that the store locators they studied provided. These were displaying the postal address, displaying the contact information, displaying the store hours, displaying other store services, and finally a route planner. It is interesting to note the percentage of store locators that provided a route planner based on category. For apparel-foot, this was 36.4 percent. For drug pharmacy, this was 28.6 percent. For electronics, this was 50 percent. For home improvement, this was 50 percent. The overall mean was 40.5 percent. This illustrates a big opportunity for the capstone's

project store locator software to shine, as it will include driving directions, aka, route planning. It will also be sure to incorporate the rest of the features.

According to Wandosell et al, “...most large scale retail firms included store locator tools on their websites, but the type and format of presentation varied significantly among them... most websites included interactive web maps...but some retail websites still included this information using non-interactive tools...” (Wandosell et al., n.d., 11). This is very important to note because it identifies that it is vital for the software to be interactive and allow exploration by the user. Any retailer that is using a static map for a store locator is utilizing very outdated software, and given that the mean of store locators that provide route planning is only 40.5 percent, it is easy to see how the capstone is poised to set itself apart from the competition. Wandosell, et al, conclude that it is recommended that retailers use interactive maps in the store locator software. This allows the businesses to create the best connection with the consumer.

Outside of the features that the store locator will provide, there is also the added benefit of ease of use. It is targeting the developers of businesses who will be doing the actual development of their online web presence. Not only will the store locator have all of the latest features that are expected, but it will also cut development costs and time because it will eliminate the need of a developer to create their store locator from scratch. They will just be able to leverage the npm package.

Stakeholders and Community

The main stakeholders in this project are the businesses that will be opting to use the capstone project’s software components on their sites. The community that this project will benefit will be the community of developers who will be able to download and use the software component in their development cycle in order to save development time. Business owners that

are any form of brick and mortar retailer can stand to gain from having an online store locator, particularly the ones that currently have outdated store locator software. As shown by the statistics, not everyone who currently has a store locator has one that utilizes the most up to date interactive maps and route planning. This is a problem. They are missing out on a fantastic way to gain engagement with their customers. Should they fail to upgrade/update their software to the modern standard, then they will miss out on a lot of potential customers. And that is just the businesses who are using old static maps. There are retailers who merely have their addresses listed, or are relying solely on search engines.

If a company has multiple locations, then they need to have interactive maps and routes to ferry the customers into their stores. The store locator acts as a call to action to the customer. When the store locator is seamlessly incorporated into their websites, the customers enjoy the greatest ease of use. Mobile users are able to effortlessly get driving directions. All of these should be the standard today for businesses. In short, these stakeholders stand to gain or lose potential customers from utilizing this project in their online presence.

As for the community of developers, they stand to gain ease of use. Like earlier described in the introduction to this project, easy to use software can be a lifesaver. Not only that, but easy to use software is also more likely to be used widespread. The community of developers who use the software will be pleasantly surprised by the ease of use for a high quality product. They will gain saved development time.

Approach/Methodology

The approach/methodology that the student has been using for this project has already begun. Discussions with other Software Engineers over the benefits of developing the software,

and the various ways that could be employed to go about creating it have already begun. At this point in time, research is vital. There will be continued research about the best way to provide the highest quality solution before development begins. This research will include testing out software components that can be downloaded and used. As of today, this has led in the direction of the npm package manager and software registry. It is the logical choice, since it is an open source way to share JavaScript code. It is used by more than 11 million developers worldwide. The Software Engineers that have been met with have brought this up as a likely option.

That leads to the next logical steps. If npm is indeed the route that is chosen to go, then it will need to be learned. This is both exciting and stressful. Obviously there is a time constraint and something will have to be delivered ready to be tested by week five. With that in mind, there must be an approach of learning and application. There are many online resources for learning how to create node packages, and they will have to be leveraged soon. This will involve watching videos, reading articles, and obviously coding.

Once a satisfactory knowledge level of the technologies has been reached, the next step is to begin outlining the application itself. Benchmarking other software is vitally important to developing good software, and as such the very first step is to benchmark other store locators. Obviously, this has already been done to an extent in preparing this proposal. It is known what features other store locators have and what features they are missing, so the next step is to benchmark how they look. Once an idea of what a good looking store locator looks like has been developed, mocking up the layout can begin. At that point, it will be time to architect the code of the application. There are many online resources that can be used for bootstrapping an application. Once the project hits that point, it will be time to develop and get into the nitty gritty

of coding everything out. This will then be followed by testing until satisfied with the final product, and then finally the deployment of the software.

To briefly outline the approach I will be taking from start to finish:

- Research the technology that I will use to develop the application
- Learn the technology stack by using online tutorials and videos
- Benchmark other store locators
- Design the layout of the application
- Bootstrap the application;
- Code the application
- Test
- Deploy the final application.

PART 2

Ethical and Legal Considerations

Ethical Considerations

There are definitely certain ethical considerations to take into account when discussing this capstone project. The store locator project is going to use certain location based services. In order to provide route planning and directions to stores, the app will need to utilize the location of the end user. This means there is a serious privacy concern for the application's end users.

Consumers have a lot of concerns with the way businesses behave. They are very interested in making sure that businesses are conducting themselves in an ethical manner. That means that when it comes to user privacy, it will be very important to conduct development in a manner that reaches the gold standard of ethical considerations. For example, location services

are provided in many Android applications. To protect user privacy, all applications that utilize location services must first request permissions from the user. HTML Geolocation API provides the same functionality, and applications must request permissions from the user.

This is the main consideration when it comes to the users' privacy. There are very real possible negative impacts of violating the privacy of the users. Very real steps need to be taken to ensure that these possible negative impacts are mitigated. This means following all best practices when it comes to requesting permissions for a user's location. This allows the user the option to deny access to their current location.

In addition to the request for permissions, once the user's location permission is granted, it is important that the application does not betray the user's trust by using their location for something other than finding local stores and providing driving instructions. It is entirely feasible for the application to use the location for another reason. This is where being ethically sound becomes important. Again, consumers are increasingly aware of the ethical behavior of businesses. They also are aware of what companies they provide permissions to. Should they provide a location permission and then find that their information has been otherwise compromised or sold, they certainly will not be pleased with the outcome. In the end, it is desired that the project is developed with the utmost consideration of the ethical use of the end user's privacy. Their locations will not be stored or sold to anyone. The user will be satisfied knowing that their privacy is respected.

Legal Considerations

There are also certain legal considerations to take into account when discussing my capstone. Should the decision be made for the software being developed to be sold as a commercial product, there would have to be a clear outline of what rights the purchaser has.

When it comes to software, there is usually open source software and closed source software. Should the software be sold, it will have to be made clear whether or not the client will be able to make modifications to the software. If it were to be made available as open source software, then those rights would obviously be greatly expanded. There must be a careful consideration into whether or not the software will be open source or not.

In the case that the software will be made to be sold, there will be the considerations for when and how payment will be made. Will the payment be up front or will it be subscription based? In the case of subscription based, will the use of the software be terminated when payment stops? The biggest legal consideration for this project is whether or not the software will be free or will be paid for.

There are also the issue of warranties and disclaimers. Will the software come with any sort of warranty? This is a big consideration when the decision is made whether this product will be free or paid. If it is paid, there will almost certainly need to be some sort of assurance that the software will work. In the case that the software is free, then it will be less of a necessity to come with a warranty, especially if it is made to be open source software. Clients will be able to modify the source code and fix any problems or add in any missing features if it is made free. Ultimately the amount of assurances that need to be made over the functionality of the software comes down to the decision if it will be free and open source or paid and closed source.

Due to the scope of the project and the limited time for which the capstone can be completed, it is highly likely that this software will go the route of free to use, open source software. With this route, there will not necessarily need to be a warranty assuring that the software is in working order. However, it will be important for the software to be licensed correctly as open source software so that it can be shared and modified by anyone. An example

of this would be the MIT License. As it comes to legal considerations, this is probably the easiest and simplest route to take. The MIT license comes with a clause that states the software is provided “As Is”, so without warranty. It also holds the authors free from liabilities. Though the decision is not currently final, due to the ease of use and stress free way it navigates the legal aspects of this project, it is the likely choice at the present time.

Part III

PROJECT SCOPE

Timeline / Budget

A timeline for the project is listed below:

Week	Task
0	Research Angular + NPM
1	Benchmark Current Online Store Locators
2	Design Layout of Application
3	Code Interactive Map
4	Code Search Functionality
5	Testing + Focus Group
6	Finished Coding
7	Capstone Event

The main goal of the timeline is to keep on schedule for the testing + focus group that will need to be done by the end of week 5. In order to have meaningful feedback, the majority of the work will need to be done so that the focus group can spend time with a build of the application that showcases most of, if not all of, the functionality of the application. There will

also need to be enough time to implement any changes based upon the feedback from the focus group before the capstone event in week 7.

There is no budget for the project, and it will leverage free resources for development and deployment. Github will be used for the repo purposes with a free public repository, and the current options for deploying test applications will be using free heroku deployment with a free JawsDB database if this is needed. The deployment process with an NPM package should also be free.

Resources Needed

The resources needed for this project are the development computer used by the student as well as the web development software that will be used. This includes technology such as Javascript, Angular, and Node.JS. The IDE's that will be used will most likely include WebStorm and/or Visual Studio for the development of HTML and Typescript pages.

In addition to the development software that will be used, there are also many online resources that will need to be referenced for the development and deployment of the software, particularly in the realm of NPM packages. These online resources involve the proper usage of Node.Js, Angular, and Google Maps API. Google Maps API will be the primary resource for the development of the interactive maps. The online documentation of the api will be a heavily referenced resource for this project.

Milestones

The main milestones of the project are listed in the timeline of the project above. These are the benchmarking of other store locators, finishing the coding of the interactive map, coding

the search functionality for the stores, coding the handling of the data sources, the testing, and finally the deployment.

After seeking feedback in the testing milestone, then there will be another milestone implementing any changes based upon the feedback. It will be important to note the feedback and make any necessary changes should there be issues that are brought up by the testers.

Risks and Dependencies

One of the major risks in this project is that the student has never deployed a package to NPM before. Since this is the main way that the project will be deployed, and the core idea behind the project is to create reusable components that can be used by other developers to create Store Locators, there is a major risk of problems during both the development and deployment process.

Ultimately, this is believed to be an acceptable risk because there is sufficient experience in working with Web Development and Node.js such that it is not expected for there to be a big jump in knowledge required to complete the project.

Final Deliverables

There will be two main final deliverables for this project. The first deliverable will be an NPM package deployed that can be downloaded, installed, and used in Node.js applications. The second deliverable will be a working example of the Store Locator application working with sample data that will be deployed most likely to a test environment on Heroku.

Usability Testing and Evaluation

Usability testing will be done with members of a focus group. These members will include both technical and non-technical members. In order to complete the focus group testing, a working prototype will be deployed to a test environment, probably on Heroku, where the testers can navigate to the site and test all of the functionality. They will be asked to evaluate the workflow of the store locator. This involves viewing the different ways the user can search for a store: by current location, by state, by city, by zipcode, etc. They will also be asked to evaluate the visual look and feel of the application.

For the technical users, there will be documentation provided detailing how the component will be used in development. They will be asked to evaluate the ease of use of the component. If time permits, it will be the goal to allow the technical testers to download and use the component on their own. However, if this is not possible at the 5 week mark, then feedback on the core functionality and look of the application will suffice.

Bibliography

Wandosell, G., Parra-Merono, M. C., & Banos, R. (n.d.). Online Store Locator: An Essential Resource for Retailers in the 21st Century.